Gaurav Vilas Chaudhari

AI/ML Engineer | LLMs, RAG, Agentic AI and Computer Vision

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SUMMARY

AI/ML Engineer at Stark Digital building production LLM systems for government and enterprise clients: retrieval-augmented generation (RAG), natural-language-to-SQL analytics, LangGraph multi-agent workflows and real-time computer vision, served with FastAPI, PostgreSQL/pgvector, Celery and Redis and deployed on edge and mobile devices.

TECHNICAL SKILLS

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Orchestration, LangGraph, LangChain, Multi-Agent Systems, Model Context Protocol (MCP), Text-to-SQL, Embeddings, Hybrid Search, HyDE, Gemini

Machine Learning & Vision: PyTorch, TensorFlow, scikit-learn, YOLOv8, OpenCV, OCR, NLP, SHAP

Backend & Data: FastAPI, REST APIs, WebSockets, PostgreSQL, pgvector, Vector Search, Redis, Celery, Distributed Inference

MLOps & Tools: Docker, Git, GitHub, LangSmith

Languages & Frameworks: Python, C++, Java, JavaScript, TypeScript, SQL, React, Next.js, React Native, Flutter

EXPERIENCE

AI Engineer | Stark Digital, Pune **Jun 2026 – Present**

- Built a natural-language-to-SQL analytics assistant (FastAPI, Vanna 2.0, PostgreSQL) for PMRDA and PMC government databases, with role-based access, a read-only SQL guard and questions in English, Marathi and Hinglish.
- Architected a multi-tenant RAG document search platform (DMS) with hybrid pgvector and keyword search, a HyDE fallback, multilingual query expansion and asynchronous ingestion on Celery and Redis.
- Developed Project Iris, a multi-camera occupancy system using YOLOv8 and OpenCV that streams telemetry over WebSockets to IoT controllers, switching lighting and HVAC by zone at about 1 s latency.
- Delivered an edge-AI pothole detection app for MIDC (React Native, FastAPI) that combines YOLO detection with Gemini multimodal analysis for severity scoring and false-positive filtering.
- Built a LangGraph multi-agent intent router with a pgvector semantic classifier that auto-answers FAQs and escalates investment proposals to officers.

Engineering Apprentice | CMPDI RI-4, Nagpur · Certificate **Feb 2025 – Feb 2026**

- Engineered backend automation workflows and SQL data-access layers for enterprise web applications.
- Built the Vehicle Booking System (PHP, SQL) with login, availability checks and an admin approval panel.

Python Intern | MDB Electrosoft Pvt. Ltd., Amravati · Certificate **Jul 2022 – Aug 2022**

- Completed a six-week internship in Python programming and GUI development, delivering the assigned projects on time.

PROJECTS

- **Pustakosh: AI Document Library** *(FastAPI, PostgreSQL/pgvector, Celery, Next.js)*: hybrid search with reranking, multilingual query expansion, page-cited answers and per-tenant row-level security.
- **Multi-Agent PMC Officer Assistant** *(FastMCP, OpenRouter, PostgreSQL)*: five-agent orchestrator with a guarded MCP tool server (row limits, timeouts, no writes) and e5-embedding template retrieval with reciprocal rank fusion.
- **AI FinTech Auto-Reconciler** *(React Native, Kotlin, C++/JSI)*: on-device receipt reading with Gemini Nano and ML Kit OCR, and three-tier transaction matching through a custom JSI bridge.
- **Credit-Card Fraud Detection** *(scikit-learn, SHAP, FastAPI)*: Random Forest classifier with ROC-AUC 0.98 on held-out data, a cost-weighted decision threshold and SHAP explanations.

EDUCATION

B.E. in Computer Science and Engineering | SGPA 8.6 **2023 – 2026**

Shri Sant Gajanan Maharaj College of Engineering (SSGMCE), Shegaon

Diploma in Computer Engineering | 87.09% **2020 – 2023**

Government Polytechnic, Achalpur

CERTIFICATIONS

- Introduction to Large Language Models (LLMs) | NPTEL, IIT Madras
- Deep Learning | NPTEL, IIT Ropar
- Data Structure and Algorithms using Java | NPTEL, IIT Kharagpur
- Machine Learning with Python | IBM (Coursera)
- Introduction to Git and GitHub | Google (Coursera)
- Building Web Applications in PHP | University of Michigan (Coursera)